

Source Code

LAB 3.1 Implement Program that identifies valid c identifiers.

```
c_keywords = {"auto", "break", "case", "char", "const", "continue", "default", "do",  
"double", "else", "enum", "extern", "float", "for", "goto", "if", "int", "long", "register",  
"return", "short", "signed", "sizeof", "static", "struct", "switch", "typedef", "union",  
"unsigned", "void", "volatile", "while"}
```

```
str = input("Enter an identifier to check: ")
```

```
if len(str) == 0:
```

```
    print("Invalid Identifier")
```

```
elif not (str[0].isalpha() or str[0] == "_"):
```

```
    print("Invalid Identifier")
```

```
elif not all(ch.isalnum() or ch == "_" for ch in str):
```

```
    print("Invalid Identifier")
```

```
elif str in c_keywords:
```

```
    print("It is a C keyword")
```

```
else:
```

```
    print("Valid identifier")
```

Output

```
Enter an identifier to check: int
It is a C keyword
○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

```
Enter an identifier to check: _1abc
Valid identifier
○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

Conclusion

The program successfully identifies whether an input string is a valid C identifier. It checks starting characters, allowed characters, and reserved C keywords. Thus, the basic rules for C identifier validation are implemented successfully.